

- 57) What term describes infectious diseases that are native to animals but can be transmitted to humans?
- A. Zoonoses
 - B. Vector-borne diseases
 - C. Nosocomial infections
 - D. Foodborne diseases
- 58) Which of the following diseases is not considered a zoonosis?
- A. Rabies
 - B. SARS-CoV-2
 - C. Human immunodeficiency virus (HIV)
 - D. Tetanus
- 59) What are reemerging diseases?
- A. Newly discovered infectious diseases increasing in frequency.
 - B. Older, well-known diseases that are increasing in occurrence.
 - C. Diseases confined to specific regions.
 - D. Non-infectious diseases that affect large populations.
- 60) Which of the following is an example of a reemerging infectious disease?
- A. SARS-CoV-2
 - B. Tuberculosis
 - C. Ebola fever
 - D. Zika virus infection
- 61) What is a fundamental reason for the persistence and reemergence of infectious diseases?
- A. Microbial adaptation to changes in individuals, communities, and environments
 - B. Lack of new antibiotics
 - C. Overuse of vaccines in the global population
 - D. Limited migration of populations
- 62) Which of the following is not a common reemerging infectious disease?
- A. Influenza
 - B. Malaria
 - C. Cholera
 - D. Polio
- 63) What is a clinical specimen?
- A. A piece of medical equipment used for diagnosis
 - B. A biological material taken from a patient for testing or examination
 - C. A synthetic sample used for laboratory training
 - D. An environmental sample collected for research purposes
- 64) What is the first step in the laboratory diagnosis of an infectious disease?
- A. Identifying the causative microbe using culture techniques
 - B. Collecting a clinical specimen from the patient
 - C. Administering antibiotics empirically
 - D. Performing a biochemical test on the patient's blood

- 65) Why is proper collection and handling of clinical specimens critical in laboratory diagnosis?
- A. To save time during processing
 - B. To ensure the accuracy of test results and avoid contamination
 - C. To reduce the cost of laboratory equipment
 - D. To eliminate the need for multiple diagnostic tests
- 66) What is antagonism in a microbial community?
- A. A mutualistic relationship where microbes benefit each other
 - B. A competition for space and nutrients in a shared habitat
 - C. A symbiotic relationship with no impact on any organism
 - D. A cooperative interaction where microbes share nutrients
- 67) Which of the following methods of infection control requires the use of an autoclave?
- A. sanitization
 - B. sterilization
 - C. irradiation
 - D. disinfection
- 68) Which of the following was the first widely used antiseptic and disinfectant?
- A. Iodine
 - B. Alcohol
 - C. Merthiolate
 - D. Chlorine
 - E. Phenol
- 69) The main feature of prokaryotic organisms is
- A. Absence of locomotion
 - B. Absence of nuclear envelope
 - C. Absence of nuclear material
 - D. Absence of protein synthesis
- 70) Kuru disease in Humans is caused by
- A. Bacteria
 - B. Viroides
 - C. Prions
 - D. Mycoplasma
- 71) During conjugation the genetic material will be transferred through
- A. Cell wall
 - B. Medium
 - C. Pili
 - D. Capsule
- 72) Antiseptic surgery was discovered by
- A. Joseph Lister
 - B. Ernest Abbe
 - C. Pasteur
 - D. Beijerinck

- 73) Protein particles which can infect are called
- A. Virons
 - B. Prions
 - C. Nucleoids
 - D. None of these
- 74) The viruses that live as parasites on bacteria are
- A. Fungi
 - B. Commensals
 - C. Bacteriophages
 - D. None of these
- 75) The characteristics and shape of bacterial cells is maintained because of
- A. Capsule
 - B. Cell wall
 - C. Cell membrane
 - D. Slim layer
- 76) The microorganisms which are permanent residents of certain body sites are
- A. Bacteria
 - B. Fungi
 - C. Normal flora
 - D. Both b and c
- 77) An object that can harbour and transmit microorganisms:
- A. Mite
 - B. Fomite
 - C. Arthropod
 - D. Vector
- 78) Nosocomial infections are:
- A. Infections developed while the patient is in the hospital
 - B. Infections of the nasal area
 - C. Infections where pathogens enter the body through the nose
 - D. Infections among animals
- 79) Rod-shape bacteria curved from commas are known as:
- A. Bacilli
 - B. Cocci
 - C. Spiral
 - D. Vibrios
- 80) The organism with atypical cell wall:
- A. Mycoplasmas
 - B. Rickettias
 - C. Chlamydias
 - D. Viruses

- 81) Koch's postulates include the following, except:
- A. The suspected organism should be present in healthy individuals
 - B. The organism must be isolated and grown as pure culture in the laboratory
 - C. The organism must initiate the symptoms of the disease in healthy animals
 - D. The organism must be re-isolated from the animal and be cultured
- 82) A structurally mature infection viral particle:
- A. Virion
 - B. Capsid
 - C. Capsule
 - D. spore
- 83) Virus lack all of the following organelles, except:
- A. Mitochondria
 - B. Nucleus
 - C. Mitotic apparatus
 - D. DNA
 - E. Nuclear membrane
- 84) The following are true about prokaryotes, except:
- A. They are relatively small.
 - B. Their genes are dedicated to essential functions only.
 - C. They lack autonomous organelles.
 - D. Their genetic materials are enclosed in a nuclear membrane.
- 85) Unicellular prokaryotic organism that divides by binary fission:
- A. Bacteria
 - B. Fungi
 - C. Protozoa
 - D. Helminths
- 86) The bacterial organelle that is used for motility:
- A. Flagella
 - B. Pili
 - C. Cilia
 - D. Pseudopods
- 87) This mediates the attachment of bacteria to specific receptors on human cell surfaces, which is the necessary step in the initiation of infection for some organisms:
- A. Flagella
 - B. Fimbriae
 - C. Capsule
 - D. Endospores
- 88) This mediates the firm adherence of bacteria to various structures such as skin, heart valves, teeth and even catheters:
- A. Capsule
 - B. Slime layer
 - C. Pili
 - D. Flagella

- 89) It is the relationship between the host and normal microbiota:
- A. Disease
 - B. Infection
 - C. Symbiosis
 - D. Parasitism
- 90) It refers to the invasion or colonization of the body by pathogenic bacteria:
- A. Disease
 - B. Infestation
 - C. Infection
 - D. Syndromes
- 91) The following structures normally contain Indigenous microbial flora, except:
- A. Teeth
 - B. Urethra
 - C. Bronchi
 - D. Skin
- 92) The region of the body that contains the largest population of microbial flora:
- A. Skin
 - B. Colon
 - C. Vagina
 - D. Mouth
- 93) What is antibiosis?
- A. A process where microbes use up all nutrients in the environment
 - B. An interaction where metabolic products of one organism harm or inhibit another
 - C. A neutral interaction between microbes in a shared space
 - D. A process where microbes form mutual benefits for survival
- 94) What is normal flora?
- A. Microorganisms that always cause disease in the host
 - B. Microorganisms that frequently colonize a specific site in healthy individuals
 - C. Microorganisms that only appear during infections
 - D. Pathogenic microbes that remain dormant in the body
- 95) What is another term used to describe normal flora?
- A. Pathogens
 - B. Opportunists
 - C. Indigenous microflora
 - D. Foreign microbiota
- 96) What distinguishes transient microbes from resident microbes?
- A. Transients are permanently established, while residents are temporary.
 - B. Transients occupy the body temporarily, while residents become established.
 - C. Transients cause disease, while residents are always harmless.
 - D. Transients are only found during infections, while residents are normal flora.

- 97) How does the normal flora avoid being targeted by the body's defenses?
- A. By multiplying faster than the immune system can respond
 - B. By producing toxic substances that suppress immune cells
 - C. By adapting to avoid detection by the host's immune defenses
 - D. By hiding inside host cells
- 98) Which of the following factors influences the distribution and composition of the normal microbiota?
- A. Nutrients
 - B. Physical and chemical factors
 - C. Host defenses
 - D. All of the above
- 99) How do mechanical factors influence the microbiota in the oral cavity?
- A. altering the pH of the saliva
 - B. dislodging microbes during chewing and tongue movements
 - C. neutralizing the immune response
 - D. increasing microbial growth
- 100) Which of the following conditions is least likely to cause fluctuations in the normal microbiota?
- A. Stress
 - B. Hormonal changes
 - C. Consistent diet
 - D. Drug therapy
- 101) What is one key way the microbiota benefits the human host?
- A. causing mild infections to boost immunity
 - B. preventing the colonization of harmful microorganisms
 - C. directly destroying host tissues for nutrient release
 - D. suppressing the production of vitamins
- 102) What is microbial antagonism?
- A. The harmful effects of pathogens on the human body
 - B. The antagonistic effect of "good" microbes against harmful intruder microorganisms
 - C. The competition among harmful pathogens for nutrients
 - D. The neutral relationship between microbes and the host
- 103) Which vitamin is synthesized and excreted by the microbiota in the human gut?
- A. Vitamin D
 - B. Vitamin K
 - C. Vitamin A
 - D. Vitamin E
- 104) Which of the following best describes microorganisms?
- A. Large, multicellular organisms with complex structures
 - B. Microscopic, relatively simple organisms that are usually unicellular
 - C. Visible organisms that can only be found in water environments
 - D. Organisms that do not require any genetic material for reproduction

- 105) A 22-year-old woman who works in a plant nursery presents with a history of fever and cough for 2 months. Over this period she has lost 5 kg. Chest radiography shows bilateral upper lobe infiltrates with cavities. A stain of her sputum shows acid-fast bacilli. The likely means by which the patient acquired her infection is
- A. Sexual activity
 - B. Ingesting the microorganisms in her food
 - C. Holding onto contaminated hand rails when she takes public transportation
 - D. Handling potting soil
 - E. Breathing aerosolized droplets containing the microorganism
- 106) A 65-year-old woman has a long-term central venous catheter for intravenous therapy. She develops fever and subsequently has multiple blood cultures positive for *Staphylococcus epidermidis*. All of the *S. epidermidis* isolates have the same colony morphology and antimicrobial susceptibility pattern, suggesting that they are the same strain. An *S. epidermidis* biofilm is thought to be present on the catheter. Which one of the following statements about such an infection is correct?
- A. The biofilm containing the *S. epidermidis* is likely to wash off the catheter.
 - B. Production of an extracellular polysaccharide inhibits growth of the *S. epidermidis*, limiting the infection.
 - C. The *S. epidermidis* in the biofilm are likely to be more susceptible to antimicrobial therapy because the bacteria have decreased rates of metabolism.
 - D. The quorum-sensing ability of *S. epidermidis* results in increased susceptibility to antimicrobial therapy.
 - E. The complex molecular interactions within the biofilm make it difficult to provide effective antimicrobial therapy, and it is likely the catheter will have to be removed to cure the infection
- 107) Which method of reproduction is most commonly used by bacteria?
- A. Binary fission
 - B. Budding
 - C. Sexual reproduction
 - D. Conjugation
- 108) Bacteria are unique among microorganisms because they:
- A. Do not require a host cell for reproduction and can live freely
 - B. Must always rely on other cells to reproduce
 - C. Are exclusively multicellular organisms
 - D. Contain no genetic material in their structure
- 109) What is one reason bacteria are often described as "free-living" organisms?
- A. They lack genetic material and do not reproduce on their own.
 - B. They can survive and reproduce without needing a host organism.
 - C. They only grow in highly specific, dependent environments.
 - D. They are primarily found within host cells.
- 110) Which of the following organisms is classified as a prokaryote?
- A. Fungi
 - B. Algae with a true nucleus
 - C. Blue-green algae
 - D. Protozoa

- 111) How do bacteria and blue-green algae differ in terms of their cellular components?
- Bacteria contain chlorophyll, while blue-green algae do not.
 - Blue-green algae contain chlorophyll, while bacteria do not.
 - Both bacteria and blue-green algae lack chlorophyll.
 - Both contain chlorophyll but are unable to photosynthesize.
- 112) What distinguishes prokaryotic cells from eukaryotic cells?
- Prokaryotic cells are always multicellular, while eukaryotic cells are unicellular.
 - Prokaryotic cells lack a membrane-bound nucleus, whereas eukaryotic cells have one.
 - Prokaryotic cells have a more complex structure with numerous organelles.
 - Prokaryotic cells contain chlorophyll, unlike eukaryotic cells.
- 113) What is the smallest diameter that can typically be resolved and seen clearly with the naked eye?
- 0.2 μm
 - 2 μm
 - 200 μm
 - 2000 μm
- 114) What is the shape of bacilli?
- Spherical
 - Spiral
 - Rod-shaped (cylindrical)
 - Oval
- 115) What is the primary distinction between coccobacilli and true cocci?
- Coccobacilli are spiral-shaped, whereas cocci are rod-shaped.
 - Coccobacilli are oval, while cocci are strictly spherical.
 - Coccobacilli are larger than cocci.
 - Coccobacilli are rod-shaped, but with lengths similar to their width, unlike true cocci which are fully spherical.
- 116) Which of the following is NOT a function of the prokaryotic cell wall?
- Determining the shape of the cell
 - Protecting the cell from osmotic lysis
 - Controlling the cell's movement
 - Protecting the cell from toxic substances
- 117) Which of the following best describes a primary function of the cell wall in determining bacterial shape?
- The cell wall gives a spherical shape to all bacteria.
 - It provides a rigid structure that determines and maintains the cell's shape.
 - It allows bacteria to change shape based on their environment.
 - It stores genetic material that controls cell shape.
- 118) Which component of peptidoglycan is connected to a peptide chain?
- N-acetylglucosamine
 - N-acetylmuramic acid
 - L-lysine
 - Glucose